MIDTERM EXAM SHEET

Name:
StudentID:

Instructions
You have 90 minutes to complete the exam. Answer Parts A and B.

Part A: The multiple selection questions are worth 40% in total and each question is worth the same amount. The score of part A is calculated using the following formula:

$$\text{Max}\left(\frac{\#\text{selected correct answers} - \#\text{selected incorrect answers}}{\text{total \# of correct answers}} \times \text{total \#of points}, 0\right)$$

Use the special answer sheet for Part A.

Part B: Part B contains FOUR questions. Answer any THREE. Each is worth 20%. You may use the back of the exam sheets if you need extra space.
The exam will be closed-book. You are not allowed to use phones and laptops too. However, you are allowed to bring a hard copy of my slides with review notes (without multiple choice questions and multiple selection questions).

PART A: MULTIPLE SELECTION QUESTIONS

Question 1: What of the following strings match the regular expression $a^+(a+b)^*$

A. babab

B. abbbb

C. aaaaaa

D. ab

Question 2: Identify the language generated by the following grammar:

$$S \to XY$$
$$X \to aX \mid \varepsilon$$
$$Y \to aYb \mid \varepsilon$$

A. $\{a^m b^n \mid m \geq n,\ n > 0\}$

B. $\{a^m b^n \mid m \geq n,\ n \geq 0\}$

C. $\{a^m b^n \mid m > n,\ n \geq 0\}$

D. $\{a^m b^n \mid m > n,\ n > 0\}$

Question 3: Which of the following is true for the language $L = \{a^n b^n c^n \mid n >= 1\}$?

A. It is Regular

B. It is not Regular but Context Free

C. It is neither Regular nor Context Free

D. It is Recursively Enumerable

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Question 4: Transition function of a Deterministic Finite Automaton $D = (Q, \Sigma, \delta, q_0, F)$ maps

A. $\Sigma \times Q \to \Sigma$

B. $\Sigma \times \Sigma \to Q$

C. $Q \times Q \to \Sigma$

D. $Q \times \Sigma \to Q$

Question 5 In a context-free grammar $G = (V, \Sigma, R, S)$, V is the set of

A. Non-terminal symbols

B. Terminal symbols

C. Variables

D. Nullable variables

Question 6: Which of the following components are not in formal definition of a Nondeterministic Pushdown Automaton?

A. Transition function

B. Blank symbol

C. Set of states

D. Stack alphabet

E. Configuration

Question 7: Which of the following are correct about finite automata?

A. Finite automata cannot remember arbitrary large amount of information

B. Finite automata sometimes recognize languages that are not regular

C. There is no finite automata that can recognize set of binary strings of equal 0's and 1's.

D. Some finite automata fail to recognize regular languages

E. There is a finite automaton that recognizes the set of strings over "(" and ")" and have balanced parenthesis

Question 8: Consider the CFG G

$$S \to A \mid B$$
$$A \to aaA \mid \varepsilon$$
$$B \to bB \mid \varepsilon$$

What is correct about G?

A. All strings in G have at least one a's

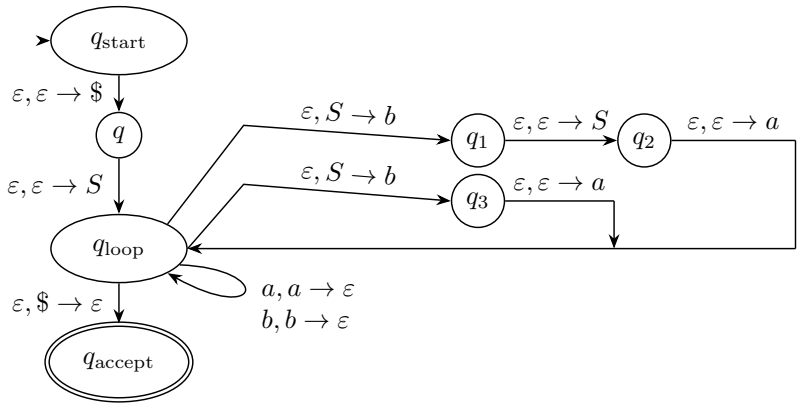
B. G is in CNF

C. G is ambiguous

D. S is nullable

E. $aabb \in L(G)$

Question 9: Given the Push Down Automaton P



Which of the following is the order of configuration (q_1, ab, bS) in the computation of P with string aaabbb. It is known that the first configuration is $(q_{\text{start}}, ab, \varepsilon)$

A. 2

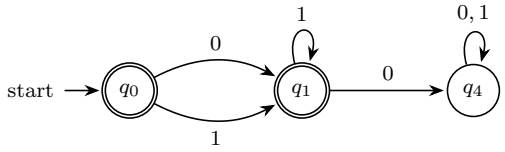
B. 3

C. 4

D. 5

E. 6

Question 10:
Consider the following DFA D



Which of the following regular expressions denote the language accepted by D?

A. $\varepsilon + 01^* + 1^*$

B. $\varepsilon + 01^* + 1^+$

C. $01^* + 1^*$

D. $01^* + 1^+$

E. $1(0+1)^* + 0(0+1)^*$

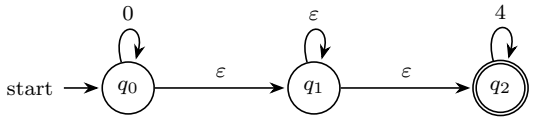
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ANSWER SHEET FOR PART A

QUESTION NUMBER	CORRECT ANSWERS
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

PART B: ANSWER QUESTIONS

1/Let N be the following nondeterministic finite automaton.



a) Write the formal definition of N

b) Convert N to the equivalent DFA: D

2/ Given the regular expression over alphabet $\{1, 2, 3\}$

$$123(1 + 2 + 3)^*$$

a) Use proper English to describe the language defined by the regular expression

b) Convert the regular expression to its equivalent NFA

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3/ Convert the following grammar to Chomsky Normal Form

$$S \to ABa$$
$$A \to aabA \mid a$$
$$B \to Ac$$

4/ Prove that language $L = \{ww^R \mid w \in \{0,1\}^*\}$ is not context free.

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